Explore core analytic skill

Let's recap what we've learned

about analytical thinking so far.

The 5 key aspects are visualization, strategy,

problem-orientation,

correlation, and using big-picture

and detail-oriented thinking.

We've seen how you already

use them in your everyday life.

We also talked about how different people

naturally use certain types of thinking,

but that you can absolutely grow and develop

the skills that might not come as easily to you.

This means you can become a versatile thinker,

which is a very important part of data analysis.

You might naturally be an analytical thinker,

but you can learn to think creatively

and critically, and be great at all three.

The more ways you can think,

the easier it is to think outside

the box and come up with fresh ideas.

But why is it important to think in different ways?

Well because in data analysis, solutions

are almost never right in front of you.

You need to think critically to

find out the right questions to ask.

But you also need to think creatively

to get new and unexpected answers.

Let's talk about some of the questions data

analysts ask when they're on the hunt for a solution.

Here's one that will come up a lot:

What is the root cause of a problem?

A root cause is the reason why a problem occurs.

If we can identify and get rid of a root cause,

we can prevent that problem from happening again.

A simple way to wrap your head around

root causes is with the process called the Five Whys.

In the Five Whys you ask

"why" five times to reveal the root cause.

The fifth and final answer should give you

some useful and sometimes surprising insights.

Here's an example of the Five Whys in action.

Let's say you wanted to make

a blueberry pie but couldn't find any blueberries.

You've been trying to solve a problem by asking,

why can't I make a blueberry pie?

The answer will be, there

are no blueberries at the store.

There's Why Number 1.

You then ask, why were there no blueberries at the store?

Then you discover that the blueberry bushes

don't have enough fruit this season.

That's Why Number 2.

Next, you'd ask, why was there not enough fruit?

This would lead to the fact that

birds were eating all the berries.

Why Number 3, asked and answered.

Now we get to Why Number 4.

Ask why a fourth time and the answer would be that,

although the birds normally prefer

mulberries and don't eat blueberries,

the mulberry bush didn't produce fruit this season,

so the birds are eating blueberries instead.

Finally, we get to Why Number 5,

which should reveal the root cause.

A late frost damaged the mulberry bushes,

so it didn't produce any fruit.

You can't make a blueberry pie

because of the late frost months ago.

See how the Five Whys can reveal

some very surprising root causes.

This is a great trick to know, and it can be

a very helpful process in data analysis.

Another question commonly asked by data analysts is, where are the gaps in our process?

For this, many people will use

something called gap analysis.

Gap analysis lets you examine and evaluate how a process

works currently in order

to get where you want to be in the future.

Businesses conduct gap analysis

to do all kinds of things,

such as improve a product or become more efficient.

The general approach to gap analysis is

understanding where you are now

compared to where you want to be.

Then you can identify the gaps that exist between

the current and future state

and determine how to bridge them.

A third question that data analysts ask a lot is,

what did we not consider before?

This is a great way to think about what information

or procedure might be missing from a process,

so you can identify ways to make

better decisions and strategies moving forward.

These are just a few examples of the kinds of

questions data analysts use at their jobs every day.

As you begin your career,

I'm sure you'll think of a whole lot more.

The way data analysts think and ask questions plays a big part in how businesses make decisions.

That's why analytical thinking

and understanding how to ask

the right questions can have

such a huge impact on the overall success of a business.

Later, we'll talk more about how data-driven

decisions can lead to successful outcomes.